Materials and Methods

Study design and data sources

Analyses were conducted on data-complete species panels defined for each data type. The transposable-element analyses comprised 34 species, the microscopy and phylogenetic-comparative analyses comprised 24 species, and 21 species occurred in both panels. *Desmognathus brimleyorum*, *D. folkertsi*, and *D. ochrophaeus* were represented in the microscopy panel but lacked transposable-element data, and no values were imputed for these species. Genomic and microscopy measurements were joined at the species level and were not assumed to have been obtained from the same animals.

Genome assemblies and raw sequencing resources were obtained from the National Center for Biotechnology Information Assembly and Sequence Read Archive databases. Accession-specific identity records were used to reconcile source names with the analysis taxonomy. Two dnaPipeTE outputs were available for *D. orestes*. These represented technical runs rather than biological replicates, so each run was normalized separately and the two compositions were averaged with equal weight.

Maximum snout–vent length, body-size, and life-history data were provided by Alex Pyron. Maximum SVL was used rather than total length; for *D. orestes*, the larger of two reported values (52 and 50 mm) was retained. Reproductive strategy was categorized as direct development, aquatic eggs, or terrestrial eggs with aquatic larvae. The TE panel comprised three direct developers, seven aquatic-egg species, and 24 terrestrial-egg species with aquatic larvae. Strategy assignments were supplied after inspection of the initial PCA, so strategy comparisons were exploratory. Three microscopy species lacked a frozen egg-placement assignment but retained their known aquatic-larval development and remained in the comparative fits.

Genomic read processing and de novo TE library construction

Mitochondrial reads were removed by indexing a mitochondrial reference (GenBank AY728227.1), aligning paired reads with BWA-MEM using eight threads, converting and sorting the alignments with SAMtools, and applying samtools view -f 12 -F 256 to retain pairs for which both mates were unmapped to the mitochondrial reference. These nonmapping pairs formed the nuclear-read input for subsequent TE analyses.

A *Desmognathus*-specific repeat library was generated with RepeatModeler. The workflow used the TEtools 1.87.sif container under Singularity 3.7.4, built a database for each assembly, and ran RepeatModeler with 32 threads, -LTRStruct, and -genomeSampleSizeMax 10000000000. Redundant consensus sequences were clustered with CD-HIT-EST using -d 0 -aS 0.8 -c 0.8 -G 0 -g 1 -b 500 -T 64 -M 500000. Subsequent dnaPipeTE analyses used the custom library as its TE database.

TE abundance estimation and classification

Transposable-element abundance was estimated from the nuclear-filtered short reads with dnaPipeTE. The workflow used Singularity 4.1.2, 16 CPUs, mate 1 of the nuclear-filtered reads, a configured 15-Gb genome-size denominator, 0.1× target coverage, RM_t = 0.15, two internal samples, and custom TE library. The 15-Gb value was used only to configure read sampling and was not interpreted as a measured genome size or absolute TE load.

The dnaPipeTE _reads_per_component_and_annotation output was parsed to assign repeat components to class, order, and superfamily. Within each taxonomic level, aligned bases assigned to each classified

TE category were divided by the total classified aligned bases for that species. Total aligned-base mass, retained classified mass, and unresolved mass were preserved as quality-control quantities. Unresolved mass did not enter the diversity calculations. RepeatMasker .align files used for divergence profiles described dnaPipeTE Trinity repeat-contig assemblies rather than the accession-linked genome assemblies.

TE diversity and compositional ordination

TE diversity was calculated at the superfamily level following the top-ten scope used by Wang et al. (2025). For each species, the ten most abundant strictly positive TE superfamilies were selected independently; abundance ties were resolved by superfamily name. The selected abundances were reclosed to sum to one before natural-log Shannon entropy, $H' = -\Sigma\, p_i \ln(p_i)$, and the Gini-Simpson index, $1 - \Sigma\, p_i^2$, were calculated, where p_i is the reclosed proportion of classified aligned bases assigned to superfamily i. Wang et al. (2025) do not state whether their retained top-ten abundances were reclosed; reclosure is therefore specified explicitly here.

PCA was conducted on the classified superfamily compositions of the 34 TE species. After technical-run averaging, superfamilies with positive abundance in every species were retained, leaving 24 shared superfamilies. Ginger, Merlin, Dada, Chapaev, and CR1 were excluded from the ordination because they were not detected in every species. Each retained composition was reclosed to sum to one and transformed by the centered log ratio, $\mathrm{CLR}(p_i) = \ln(p_i) - \mathrm{mean}[\ln(p)]$. The retained matrix contained no zeros, so no zero replacement was applied. CLR columns were centered without variance scaling, and principal components were obtained by singular-value decomposition. Axis signs were oriented deterministically so that the loading with the largest absolute value on each axis was positive. This ordination was retained as a supporting description of compositional variation, separate from the eight main figures.

Each diversity index was regressed separately on conditional genome size across the exact 21-species overlap. Pagel-lambda phylogenetic generalized least squares used untransformed diversity and genome size in pg/1C, with lambda estimated by maximum likelihood between zero and one. Two-sided coefficient tests and 95% intervals used the *t* distribution with 19 residual degrees of freedom. The two indices summarize the same selected composition and were not independent replications. Genome-size measurement intervals were displayed but were not propagated into these regressions.

Repeat divergence profiles

Repeat divergence profiles were reconstructed from the RepeatMasker .align files associated with the dnaPipeTE Trinity repeat-contig assemblies. Inclusive aligned hit length was calculated as |query end - query start| + 1. Rows with invalid divergence or length values were removed, and reported divergence was assigned to one-percentage-point bins with a terminal 50% bin. Within each species, aligned bases in each TE category and divergence bin were divided by the total inclusive aligned RepeatMasker hit bases for that species. Missing species-by-category-by-bin combinations were filled with zero before equal-weight averaging across the 34 species.

These fractions describe the composition of aligned repeat annotations rather than genome occupancy, insertion counts, insertion ages, or evolutionary rates. Unclassified repeats were retained as an annotation category rather than treated as a biological TE order.

A low-divergence LTR score was calculated for each species as LTR aligned-hit bases in divergence bins 0–4 divided by all LTR aligned-hit bases across all available bins. This fixed bin definition was not selected through a threshold search. An alternative denominator used all RepeatMasker aligned-hit bases. The alternative fraction equals the within-LTR score multiplied by the overall LTR fraction of aligned-hit bases, separating relative divergence-profile shape from total LTR representation. Both fractions were strictly between zero and one and were logit-transformed without a pseudocount.

For the 34 TE species, Pagel-lambda regressions compared the within-LTR score with reproductive strategy, log10 maximum SVL, and both predictors together, with an intercept-only model as a reference. Lambda and residual variance were estimated separately by maximum likelihood for each equation. Omnibus strategy tests compared models differing by two group parameters using an approximate chi-square likelihood-ratio test with two degrees of freedom. Contrasts used aquatic eggs as the reference and were reported on the raw logit scale with residual-degrees-of-freedom *t* intervals. Group prediction curves were limited to each group's observed SVL range. Group sizes did not represent independent evolutionary transitions, and post-hoc strategy coding and phylogenetic clustering limited interpretation.

Two predefined descriptive quality checks used unadjusted Spearman correlations of the within-LTR score with the dnaPipeTE unresolved fraction and the repeat-aligned fraction of its configured read sample. These cross-pipeline quantities were quality indicators rather than corrections for genome-wide TE abundance.

LTR terminal-to-internal depth proxy

DANTE, LTRharvest, and LTRdigest were used to identify candidate full-length long terminal repeat (LTR) elements and annotate their 5' LTR, internal region, and 3' LTR. Candidate sequences were joined to TEsorter annotations by exact sequence identifier and coordinates. Elements were eligible when they were at least 3,000 bp long, were classified by TEsorter as Order = LTR and Superfamily = Gypsy, and contained at least five whitespace-delimited DOMAIN|MODEL annotations. Correct parsing produced 408 eligible LTR/Gypsy elements across 30 species.

Trimmed, mitochondrial-filtered paired reads from each species were mapped to the corresponding LTR contigs with Bowtie2 using -L 20 --very-sensitive-local. SAMtools was used to convert, sort, and index the alignments and to calculate per-position depth. Positions absent from the depth output were assigned a depth of zero. Mean terminal depth was calculated as the length-weighted mean across the 5' and 3' LTRs, and mean internal depth was calculated over the annotated internal region. The terminal-to-internal ratio was obtained by dividing mean terminal depth by mean internal depth. The archived records do not preserve the Bowtie2 and SAMtools versions or the policies for mapping quality, multimappers, secondary or supplementary alignments, and duplicate reads.

Before species-level summaries were calculated, an inclusive two-sided Tukey filter was applied independently within each species to the untransformed element ratios. Elements were retained when $Q1 - 1.5 \times IQR \leq ratio \leq Q3 + 1.5 \times IQR$. The quartiles and fences were calculated from the 408 eligible elements before any coverage filtering, and 381 elements passed the IQR rule. An independent coverage gate required positive depth at 80% or more of the expected positions separately in the 5' LTR, 3' LTR, and internal region; 407 elements passed this gate. The intersection of the two filters contained 380 elements across all 30 species. Species were summarized by the median ratio with 2,000 element-bootstrap replicates and central 95% percentile intervals. Reproducible bootstrap seeds were derived from the species, filter branch, and summary statistic using SHA-256.

The terminal-to-internal statistic was used as an indirect mapping and deletion-footprint proxy motivated by the hypothesis that recombination between terminal repeats can remove internal sequences. The internal region comprised the entire intervening sequence, including zero-depth positions. A ratio of one indicates equal mapped terminal and internal depth rather than a calibrated no-deletion baseline.

Nineteen species with both conditional genome size and primary LTR summaries contributed one species median each, encompassing 239 retained elements. Pagel-lambda PGLS regressed log10 median ratio on log10 genome size with lambda constrained to zero through one. Two-sided tests and 95% coefficient intervals used 17 residual degrees of freedom. Element counts were not regression weights or independent biological replicates. Brownian covariance, ordinary regression, an untransformed PGLS, and 19 leave-one-species-out fits were retained as specified sensitivities. Measurement error and tree uncertainty were not propagated. Conversion between Gbp and pg/1C changes the intercept but leaves the log–log slope and inference unchanged.

Specimen preparation, staining, and image acquisition

Each microscopy specimen identifier represented one individual animal; multiple images and measured objects from the same specimen were repeated observations rather than independent biological replicates. For specimens sampled directly, erythrocytes were collected from an incision through the heart with a pipette and deposited at the center of a microscope slide. Slides were stained with Feulgen-Schiff reagent following Hardie et al. (2002). All images retained for morphometry and IOD were acquired by brightfield microscopy on an Olympus APX100 with a 40× objective. Slides were also reimaged with a 40× oil-immersion objective, but those images were excluded because their quality was poor.

Cell and nucleus segmentation and morphometry

Cell boundaries were segmented with Cellpose and nuclei were segmented with a custom trained YOLO model. Masks retained source-image, tile, centroid, overlap, and object-level identifiers. Objects were eligible when they formed a valid one-to-one cell-nucleus pair, did not touch an image edge, were not overlapping other cells or debris, and passed manual review.

Eligible cells were ranked within species by descending cell area. The target was the 50 largest accepted cells per species, and rejected objects were replaced by the next-largest eligible accepted cell. This procedure estimates the upper tail of detected eligible erythrocyte size rather than the population-average cell. The final morphometry dataset contained 1,152 reviewed cell-nucleus pairs across 24 species and 42 animals. Twenty-two species contributed 50 cells, *D. aeneus* contributed 44, and *D. wrighti* contributed 8.

Cell and nucleus areas were calculated from mask pixels using a linear calibration of 0.12 micrometers per pixel, equivalent to 0.0144 square micrometers per pixel. Species-level cell and nucleus estimates were medians of the accepted objects. Uncertainty intervals were obtained by paired-object bootstrap resampling conditional on the observed images and animals. These intervals do not estimate population-level variation among animals.

Nuclear IOD measurement and conditional genome-size calibration

Pixel optical density was calculated as log10(background intensity / pixel intensity), and nuclear integrated optical density (IOD) was calculated by summing pixel optical density over the nuclear mask. Images were matched using log edge sharpness and log relative ring noise; IOD, darkness, nucleus area, and cell area were excluded from the matching variables. Balance was assessed with a maximum absolute

standardized mean difference threshold of 0.10 and a maximum Kolmogorov-Smirnov distance threshold of 0.25. Species that failed either threshold remained in the release and were flagged rather than removed.

The final IOD dataset contained 805 reviewed nuclei from 51 images, 50 animals, and 24 species. For each image, the median per-nucleus IOD was calculated, and the species estimate was the arithmetic mean of the image medians so that images rather than nuclei received equal weight. Uncertainty was estimated with 2,000 hierarchical-bootstrap replicates that resampled images and then nuclei within images. Dividing each species estimate and interval by the median estimate across the 24-species panel produced a relative-IOD index used to document the measurement and calibration path.

Four IOD species had one animal, and seven morphology species had one animal; these samples could not estimate among-animal variability.

Two *D. fuscus* reference slides, prepared from specimens 32469 and 32470, served as DNA-content standards and were stained in the same experimental runs as the unknowns. Conditional genome size was calculated as target genome size (pg/1C) = (target equal-image IOD / *D. fuscus* equal-image IOD) × 16.462167689 pg/1C. The reference value was obtained from the selected pooled *D. fuscus* assembly estimate as 16.1 Gbp / 0.978 Gbp per pg = 16.462167689 pg/1C (Myers et al. 2025). The computational denominator was the finalized *D. fuscus* species estimate, rather than a recovered staining-run-matched standard-slide value. The exact standard-to-unknown run map remained unresolved, as did the approximately 41% difference between the two standard-slide median IOD values. Conditional genome-size intervals divided each target bootstrap draw by the corresponding *D. fuscus* draw and multiplied by the fixed anchor. They included sampled IOD variation but excluded uncertainty in the assembly anchor and staining-run calibration. These estimates were IOD-derived, assembly-anchored species summaries rather than independently validated C-values. IOD resampling used seed 20260710 with deterministic species offsets.

Phylogenetic comparative analyses

The focal time-calibrated *Desmognathus* phylogeny was supplied by Alex Pyron. Taxon names were reconciled with the study species, exact unsuffixed representative tips were retained, and alternate tips and species without complete data were pruned with ape::drop.tip. Single-descendant internal nodes produced by pruning were collapsed, and duplicate species tips were not permitted. The primary morphology and conditional-genome-size analyses used 24 tips. Tree sensitivity was evaluated by comparing the focal tree pruned to the same 18-species taxon set with a published 18-species topology, thereby separating taxon-set effects from differences between trees.

Positive cellular traits and maximum SVL were log10-transformed. The body-size and TE-integration extensions used the full 24-species means and sample standard deviations for every cellular trait and SVL, retaining the same scaling for reduced cohorts and alternative animal summaries. Original three-trait path sensitivities standardized the traits within each fitted dataset. Pagel-lambda PGLS estimated lambda by maximum likelihood between 10^{-7} and one for the cellular, body-size, and TE-integration equations. Each species was one model observation.

The original pairwise analyses retained forward, reverse, and two-predictor equations as association sensitivities. Coefficient tests used the *t* distribution and residual degrees of freedom; the displayed cellular pairwise coefficient intervals used the 0.975 *t* quantile with 22 residual degrees of freedom. Supporting detailed three-trait regression and path-edge tables retained normal-approximation intervals, estimate ± 1.96 standard errors. Fitted bands and coefficient intervals for the retained body-size, nuclear-

allometry, TE-integration, and expanded body-role analyses used the 0.975 *t* quantile with n − p degrees of freedom, where p included the intercept. Fitted-mean bands used the full coefficient covariance matrix. Back-transformed log-scale fitted values describe conditional medians without an arithmetic-mean bias correction, and bands are coefficient intervals rather than individual prediction intervals. These intervals condition on fitted lambda and omit uncertainty in its estimation.

Nuclear–cell scaling and paired nuclear fractions

Nuclear–cell allometry was examined by regressing log10 nucleus area on log10 cell area without standardizing the slope. The proportional-scaling hypothesis was tested with (slope − 1) divided by its standard error and n − 2 residual degrees of freedom. Four fixed comparisons crossed released pooled species medians with arithmetic means of within-animal medians, using all 24 species or excluding *D. aeneus* and *D. wrighti*. This conditional regression did not correct predictor error or establish an errors-in-variables structural exponent.

Nuclear area fractions were calculated within each actual cell–nucleus pair, summarized by animal medians, and then averaged equally across animals within species. Fractions were not ratios of aggregate medians. Using seed 20260924, 1,000 hierarchical draws resampled animals within species and paired cells within each selected animal, preserving observed sample sizes. Species summaries were arithmetic means of the sampled animal medians. Each draw retained all 24 species, and 1,000 allometry refits re-estimated lambda. Central 95% ranges summarized nuclear fractions and fitted slopes as conditional measurement sensitivity, separately from the model's *t* intervals. The source cohort retained all 42 animals and 1,152 pairs; single-animal species could not contribute estimates of between-animal uncertainty.

Body size and animal-aware sensitivities

Maximum SVL was a species maximum, not the body length of a microscopy animal. Genome size, cell area, and nucleus area were each regressed on maximum SVL. Additional equations compared cell area with genome size and SVL, nucleus area with cell area and SVL, and nucleus area with genome size, cell area, and SVL. Marginal cellular equations and two-predictor nuclear equations were also retained. All equations were fitted in the 24-species panel and the specified 22-species sensitivity excluding *D. aeneus* and *D. wrighti*. Those species were both the smallest sampled species and the only direct developers in the microscopy panel, so exclusion changed body-size range, developmental composition, and taxon membership together. No species or threshold search was used, and this comparison did not distinguish those influences or test a difference between slopes.

The animal-aware body-adjusted sensitivity used medians of within-animal morphology medians. This estimand differed from the arithmetic mean of animal medians used for nuclear fractions and the allometry alternative. For IOD, image medians were averaged within each animal and animal estimates were then averaged equally within species. Conditional genome size retained the same fixed *D. fuscus* anchor, with its denominator recomputed under each aggregation or draw.

With seed 2026092409, 250 measurement draws resampled observed animals, then images within each sampled animal, then rows within each sampled image, preserving the sample size at each level. Cell and nucleus measurements remained paired. The assays were resampled independently even when animal identifiers overlapped, so cross-assay covariance was not modeled. These draws retained all 24 species and the full-panel scaling. Separate 24 leave-one-species-out fits used released summaries and pruned trees to assess species influence. These procedures evaluated the original four body-adjusted hypotheses

and did not provide resampling support for the later expanded eight-class set. SVL and calibration uncertainty remained unmodeled.

Phylogenetic path hypotheses

The original three-trait path analysis considered conditional genome size, nucleus area, and cell area. Its candidate inventory contained 25 directed acyclic graphs representing 11 Markov-equivalence classes; ten classes had testable conditional-independence claims. The saturated class had none and was excluded from the CICc ranking. Pagel-lambda models were primary, alpha was 0.05, and classes within two CICc units of the best class were considered competitive. A fully connected three-trait diagram was retained as an illustrative saturated hypothesis, separate from scored models, and was not assigned a ranked class weight.

The original sensitivity suite comprised 250 measurement-bootstrap datasets, 24 leave-one-species-out fits, and 200 published bootstrap trees on the 18-species overlap. Measurement draws resampled pooled paired cell–nucleus objects within species and independently resampled images and then nuclei for IOD, using seed 20260710 and deterministic species offsets. The *D. fuscus* calibration denominator was recomputed in each draw. Evolutionary covariance sensitivities compared Brownian motion, Pagel lambda, and Ornstein–Uhlenbeck models with fixed and random roots. Simulation calibration used seed 20260710 and 800 datasets: 100 replicates of four generating classes under independent and Brownian residual regimes. These procedures belonged to the original three-trait comparison. Multiplication of IOD by a positive calibration constant cancels after log transformation and standardization; that invariance does not validate the absolute genome-size scale.

A subsequent four-hypothesis comparison placed SVL upstream of all three cellular traits and compared a body-only fork, a genome–cell–nucleus bridge, a genome–nucleus–cell bridge, and a genome fork to cell and nucleus. Each hypothesis represented a graphical equivalence class. The original ten testable three-trait classes were also evaluated in the 24- and 22-species cohorts to separate cohort changes from body adjustment.

The expanded body-role comparison comprised eight distinct equivalence classes. Four retained the upstream-SVL hypotheses; four allowed a genome–cell–nucleus connection with SVL linked downstream to cell area, a genome–nucleus–cell–SVL chain, SVL linked to genome size, or SVL disconnected from the cellular chain. Equivalence was checked from each graph's undirected skeleton and unshielded colliders by enumerating orientations and rejecting cyclic graphs. The genome-first cell-branch representative had four equivalent acyclic orientations. These orientations shared graphical conditional independences and were not four additional fitted hypotheses.

The eight classes were fitted separately to 24 and 22 species with fixed full-panel scaling. The primary conditional-independence test order was body size, genome size, cell area, nucleus area. A prespecified reverse order, nucleus area, cell area, genome size, body size, assessed regression-direction sensitivity for pairs without a directed path; ancestral pairs retained the graph-implied test direction. Fisher's C, parameter counts, CICc, and normalized weights were calculated from the tested claims. Weights were comparable only within one cohort, test order, and candidate set. Structural equations were fitted directly with PGLS to preserve common scaling. Reversing phylogenetic regressions could change numerical results because residual covariance parameters were re-estimated. No saturated model, exhaustive four-node search, measurement bootstrap, or tree bootstrap was added to this expanded comparison. Graphical equivalence and acceptable global fit did not identify causal direction.

TE–cellular integration

A finite regression comparison used the exact 21 species with complete TE, cellular, and maximum-SVL data. The primary within-LTR score and the all-hit-denominator alternative were logit-transformed and standardized using all 34 TE species; cellular traits and SVL retained the full 24-species log10 scaling. For each endpoint, equations modeled genome size on the TE score alone and with SVL; cell area on genome size and SVL, with and without the TE score; and nucleus area on cell area, genome size, and SVL, with and without the TE score. The most complex nuclear diagnostic had four predictors plus an intercept, leaving 16 residual degrees of freedom. No preferred equation or denominator was selected by significance.

Matched 21-species cellular coefficients were compared with the existing 24-species estimates and the TE-adjusted 21-species estimates to distinguish species membership from covariate adjustment. Prediction curves for genome size held SVL at the 21-species geometric mean. These finite equations were exploratory conditional diagnostics rather than a selected TE-to-genome-to-cell causal network. Missing TE measurements were not imputed.

All displayed regression tests were unadjusted and exploratory. Pointwise model intervals excluded tree, trait-measurement, and calibration uncertainty unless a separately described measurement sensitivity addressed a specified component. The selected large-cell cohort, incomplete animal replication, source-tree provenance, and unresolved staining calibration limited biological and causal interpretation.

Software and data availability

Downstream analyses were conducted from preserved RepeatModeler, dnaPipeTE, RepeatMasker, LTR mapping, and reviewed microscopy outputs. Local processing included species-identity joins, technical-run averaging, filtering, diversity calculations, CLR-PCA, phylogenetic analyses, and figure generation. The phylogenetic analyses used R 4.4.3, ape 5.8.1, phylolm 2.6.5, and phylopath 1.3.1. The released LTR and reviewed-morphology tables retain row-level eligibility, exclusion, and quality-control fields. Other raw and high-performance-computing outputs remain in frozen archival storage. Released files are accompanied by source paths, dimensions, column definitions, and SHA-256 checksums.